

ScholarMind

Product Requirements & Launch Brief

AI tutor workspace for source-grounded study, active recall, long-form AI Notes, canvas annotation, revision planning, and secure student workflows.

25

app pages

24

API routes

20+

file/MIME types

10

OpenRouter key slots

12MB

default upload cap

1.0

public release brief

1. Executive Summary

ScholarMind is a student-first AI study companion that turns messy course material into active revision. Students work inside Studios, upload or import notes, then ask a personalised tutor, generate study tools, annotate canvas tabs, and continue progress across devices.

Positioning: source-grounded AI tutor + active recall workspace + long-form AI Notes + canvas/revision ecosystem.

2. Market Signals

Signal	Stat	Why it matters
AI in education	\$8.3B in 2025; forecast \$57.2B by 2033	Validates timing for AI tutors and personalised learning.
E-learning services	Projected \$1.485T by 2033; 19.9% CAGR	Shows broad market pull for flexible digital study.
Mobile learning	\$95.3B in 2025 to \$392.2B by 2033	Supports mobile/tablet-first study workflows.
Quizlet	60M+ monthly users	Proves massive demand for flashcards, quizzes, and active recall.

3. Product Pillars

- **Source-grounded learning:** serious answers and tools use enabled notes/sources.
- **Personalised AI tutor:** onboarding profile shapes curriculum, country, subject, exam board, tier, goal, and learning style.
- **Active revision:** AI Notes, summaries, flashcards, quizzes, exam practice, schedules, and canvas work.
- **Studio workflow:** notebook-like tabs, split view, recent chats, saved outputs, and annotations.
- **Trust and safety:** Supabase RLS, Mind Security Engine, upload checks, usage limits, privacy/terms, sitemap, and host analytics.

4. Feature Requirements

Area	Requirements
Studio	Create/rename/delete Studios; open notes and outputs as internal workspace tabs; split view via selector, right-click, or
AI Tutor	Casual chat plus source-grounded step-by-step study help; can suggest/trigger tools; screen-aware pulse when reading
AI Notes	Long textbook-style notes with sections, formulas, callouts, worked examples, practice questions, diagrams, and scienc
Quiz	Interactive multiple-choice questions, answer reveal, explanations, and results table.
Flashcards	Swipeable active-recall cards generated from enabled sources.
Exam Generator	Practice-question mode or longer mock paper with marks and mark scheme; original exam-style questions only.
Canvas	Text annotations, highlighter/draw tools, stylus support, and tutor-visible context.
Uploads	PDF, images, DOCX, PPTX, ODP, spreadsheets, CSV/TSV, markdown, JSON, XML, HTML, RTF, TXT with validation

5. Architecture & Data

ScholarMind uses Next.js App Router, TypeScript, Supabase Auth/Postgres/Storage, OpenRouter-backed AI generation, OCR/document parsing, Framer Motion, Three.js, and a modular component/API structure.

- Supabase tables: sessions, files, chat messages, reminders, onboarding, AI usage, revision plans, site events, site settings, subscriptions.
- AI provider path: OpenRouter V2 with key rotation, timeout/retry handling, source context, and reasoning exclusion.
- Security path: local suspicious request scoring, security headers, rate limits, human verification route, optional Arcjet.
- SEO path: sitemap, robots, privacy, terms, and 10 targeted SEO pages for study-related searches.

6. Competitive Differentiation

- NotebookLM-like source grounding, but aimed at exam revision and active recall.
- Medly-style step-by-step tutoring, but tied to user-uploaded notes and visible workspace context.
- Quizlet-like cards/quizzes, but generated from personal notes and connected to tutor help.
- Long-form AI Notes plus science simulations inside the same Studio.
- Canvas annotations and stylus support let students work directly on notes and outputs.

7. Go-To-Market & Advertising

Tagline: Turn your notes into a personal AI tutor.

Primary promise: Upload notes once; revise with explanations, AI Notes, quizzes, flashcards, mock exams, and schedules grounded to your material.

- Stop rereading notes. Turn them into quizzes, flashcards, and mock exams.
- Ask an AI tutor that actually knows your class notes.
- From PDF to practice questions in one Studio.
- Revise with a tutor, not a blank chatbot.
- For students drowning in notes before exams.

8. KPIs

Funnel	Metric
Activation	First note uploaded/imported; first Studio created.
Aha moment	First generated AI tool from enabled sources.
Retention	Return to same Studio within 7 days.
Learning depth	Quiz completion, flashcard flips, exam attempts, schedule completions.
Quality	Thumbs-up/down, regenerated-output rate, unreadable-source rate.
Growth	SEO impressions, landing conversion, referrals, share/export events.

9. Launch Readiness

- Passed TypeScript typecheck.
- Passed production build.
- Passed upload-format verifier.
- Passed local production AI tool verifier for Quiz, Summary, Flashcards, AI Notes, and Exam Generator.
- Remaining: signed-in browser QA and deployed Supabase sample-upload pass.

10. Sources

Grand View Research: AI in Education Market, e-learning services press release, and mobile learning statistics.

PR Newswire: Quizlet launches advanced AI-powered tools and reports over 60M monthly users.

Repository scan: app routes, API routes, Supabase schema, document parser, security engine, AI provider, and launch checklist.